

HARDWARE BUILD LOGBOOK

Custom Differential-Drive Wheeled Mobile Robot (WMR)

Project: A Lightweight Fusion-based Approach for Autonomous Indoor Boundary Mapping

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1. Objective

This logbook documents the design and construction of a custom-built wheeled mobile robot (WMR) developed for low-cost indoor boundary mapping. The platform utilizes dual optical flow odometry, a MEMS gyroscope, and ultrasonic ranging on a single onboard controller, enabling high-frequency dead reckoning and wall-following navigation under strict cost and compute constraints.

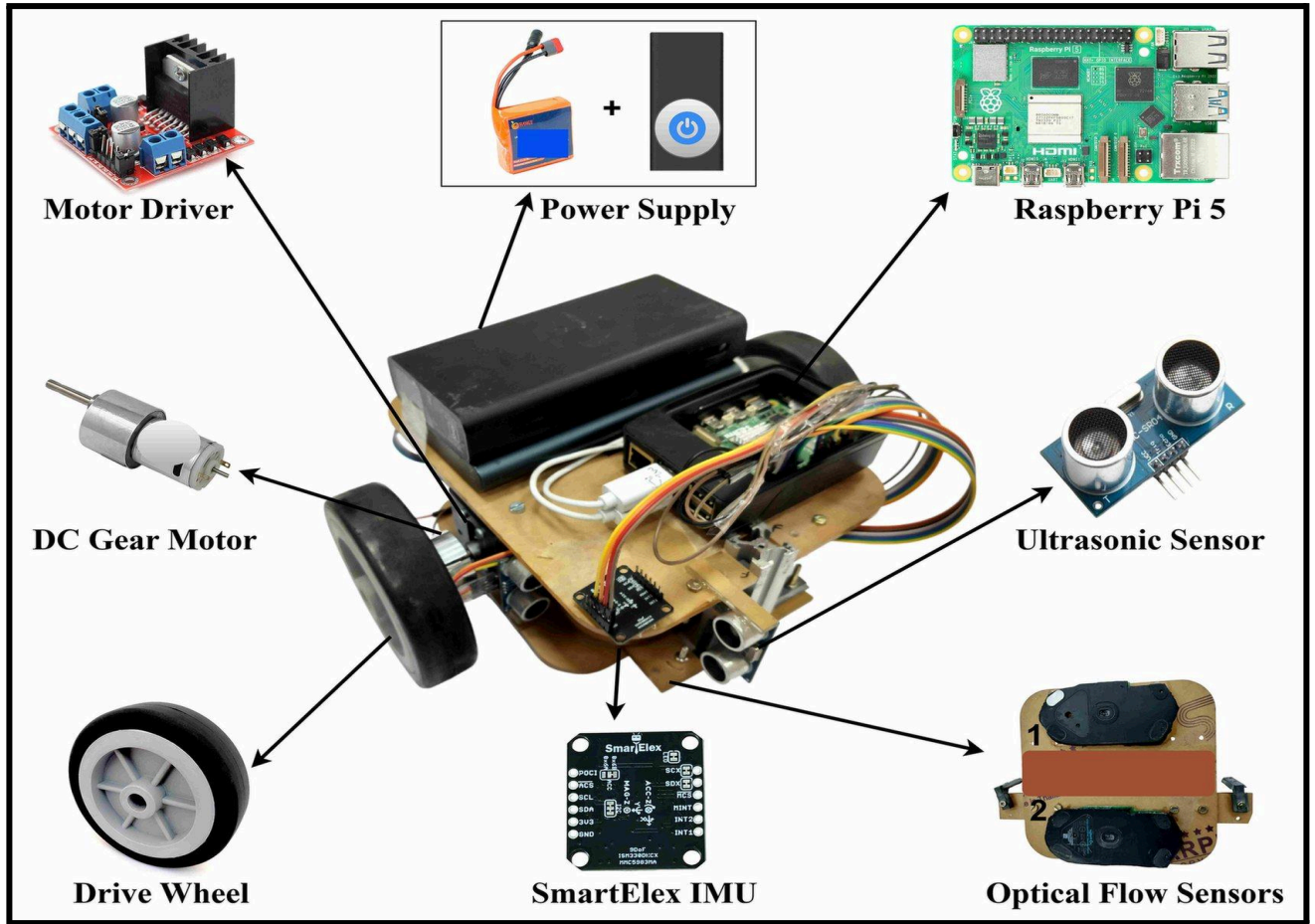


Figure 1. Custom-built WMR with all hardware components labeled.

2. Materials

Component	Specification	Qty	Function
Chassis	Acrylic plate	1	Compact mounting for all mechanical and electronic parts
Onboard computer	Raspberry Pi 5	1	Sensor polling, data logging, actuation, 100 Hz fusion
Motor driver	L298N dual H-bridge	1	Receives PWM from Pi 5, drives the two motors
Drive motors	DC gear motors	2	Differential drive, forward motion, and steering

Component	Specification	Qty	Function
Power	11.1 V Li-ion battery + switch	1	System power via integrated power switch
Optical flow	USB optical mouse sensors, 8000 DPI	2	Planar displacement (OFS1, OFS2) for dead reckoning
OFS mount	Shared acrylic plate (underside)	1	Near-constant ground contact, and baseline $d = 99.51$ mm
Gyroscope (IMU)	SmartElex ISM330DHCX	1	Continuous yaw rate is fused to reduce rotational drift
Range sensors (Ultrasonic)	HC-SR04 ultrasonic	4	Front, rear, and lateral ranging for wall following & mapping

3. Mechanical Assembly

- **Locomotion:** two DC gear motors mounted on a rigid base plate in a differential-drive configuration. Forward motion and steering are achieved through differential speed control.
- **Optical flow unit:** OFS1 and OFS2 are fixed to a shared acrylic plate on the chassis underside in near-constant ground contact. Reference points are separated by a caliper-measured baseline $d = 99.51$ mm, with the sensing x-axis aligned to the robot's forward direction.
- **Perception sensors:** The ISM330DHCX IMU is rigidly mounted on the chassis. HC-SR04 units face the front, rear, and lateral sides for wall following and boundary-point extraction.

4. Electrical and Wiring

- **Actuation:** Raspberry Pi 5 GPIO supplies PWM and direction signals to the L298N dual H-bridge, whose outputs drive the two DC motors.
- **Power:** The 11.1 V Li-ion battery powers the motor supply via an integrated switch, with a regulated 5 V rail powering the sensors, while a power bank supplies power to the Raspberry Pi 5.
- **Sensor interfaces:** OFS1/OFS2 via USB, the ISM330DHCX via I²C, and the HC-SR04 trigger/echo lines via GPIO.

5. Software, Data Logging, and Calibration

- **Fusion and logging:** the Pi 5 polls all sensors and logs timestamped data. Optical-flow displacement and gyroscopic yaw rate are fused at 100 Hz ($\Delta t = 0.01$ s). Poses and projected boundary points are stored for offline trajectory reconstruction.
- **Per-trial calibration:** OFS scale factors are obtained as $s = C_{\text{avg}} / d_s$ by translating the sensors on a uniform section of the test surface. Two fixed end points (mechanical stoppers) were positioned at a precisely measured distance (d_s) apart. The sensors were repeatedly moved between these stoppers, ensuring that each traversal covered the same known distance along a reference flat vertical surface. Multiple trials were conducted, and the corresponding sensor counts were recorded to determine the average number (C_{avg}) of units measured.
- The gyroscope bias b is estimated under stationary conditions before each run, and the OFS baseline d is measured using callipers.

6. Build Log

Build phase	Activity
Mechanical	Assembled base plate, mounted DC gear motors in differential layout, fitted drive wheels and caster, attached OFS acrylic plate to underside.
Electrical	Wired Pi 5 to L298N (PWM + direction), connected motor outputs, battery, and power switch, and verified 5 V rail and motor actuation.
Sensors	Integrated OFS1/OFS2 (USB), ISM330DHCX (I ² C), and HC-SR04 array (GPIO), and confirmed polling and 100 Hz logging.
Calibration	Measured OFS baseline (99.51 mm), computed per-axis scale factors, and estimated gyro bias under stationary conditions.